Magnetostatics

$\vec{\nabla}\cdot\vec{B}=0$, $\vec{\nabla}\times\vec{B}=\mu_0\vec{j}$

Note that $\vec{j}$ is flow of electric charge, which is discrete $\Rightarrow$ $\vec{j}$ depends on time even when the flow is steady. However magnetic field far away from the current <u>are</u> stationary. Note that $\vec{\nabla}\cdot\vec{\nabla}\times\vec{B}=0=\mu_0\underbrace{\vec{\nabla}\cdot\vec{j}}_{\text{conservation of current}}$

Landau-Lifshitz: consider $\vec{B}$ averaged over the period of charge motion.

$\vec{B}=\vec{\nabla}\times\vec{A}$ ($\vec{A}$ = vector potential) $\Rightarrow$ $\vec{\nabla}\cdot\vec{B}=\vec{\nabla}\cdot\vec{\nabla}\times\vec{A}=0$ always

$$\vec{\nabla}\times\vec{B}=\vec{\nabla}\times(\vec{\nabla}\times\vec{A})=\vec{\nabla}(\vec{\nabla}\cdot\vec{A})-\nabla^2\vec{A}=\mu_0\vec{j}$$

Since $\vec{A}$ is not unique, $\vec{B}=\vec{\nabla}\times\underbrace{(\vec{A}+\vec{\nabla}g)}_{\vec{A}'}=\vec{\nabla}\times\vec{A}$

can choose g such that $\vec{\nabla}\cdot\vec{A}'=0$ (Coulomb gauge)

Then $\nabla^2\vec{A}=-\mu_0\vec{j}$ becomes similar to

$$\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}$$

$$\Downarrow$$

$$\vec{A}(\vec{R})=\frac{\mu_0}{4\pi}\int d\vec{r}\,\frac{\vec{j}(r)}{|\vec{R}-\vec{r}|} \quad \rightleftarrows \quad \varphi(\vec{R})=\frac{1}{4\pi\varepsilon_0}\int d\vec{r}\,\frac{\rho(\vec{r})}{|\vec{R}-\vec{r}|}$$

Therefore $\vec{B}(\vec{r})=\frac{\mu_0}{4\pi}\vec{\nabla}_R\times\int d\vec{r}\,\frac{\vec{j}(r)}{|\vec{R}-\vec{r}|}$ $=\left(\vec{\nabla}\frac{1}{R}=-\frac{\vec{R}}{R^3}\right)$

$$=-\frac{\mu_0}{4\pi}\int d\vec{r}\,\frac{\vec{R}-\vec{r}}{|\vec{R}-\vec{r}|^3}\times\vec{j}(\vec{r})=+\frac{\mu_0}{4\pi}\int d\vec{r}\,\frac{\vec{j}(r)\times(\vec{R}-\vec{r})}{|\vec{R}-\vec{r}|^3}$$

(Bio-Savart law)

Contribution is max for $\vec{j}(\vec{r})\perp$ to $(\vec{R}-\vec{r})$.

$\vec{B}$ of the thin wire: $d^3\vec{r}\,\vec{j}(\vec{r}) = dV\vec{j}(r) = \text{Area}\, dl\, \vec{j} = I\, d\vec{l}$

$I = \text{Area} \cdot j$ = total current through wire's cross-section, the same everywhere

$$\Rightarrow \vec{B}(\vec{R}) = \frac{\mu_0 I}{4\pi} \int d\vec{l} \times \frac{\vec{R}-\vec{r}}{|\vec{R}-\vec{r}|^3}$$

<u>Integral form</u>

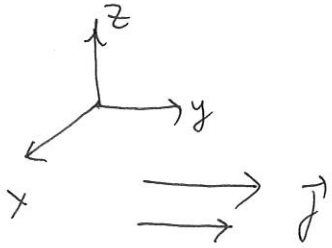
$$\oint_C \vec{B}\cdot d\vec{l} = \int_{\text{area } S} \vec{\nabla}\times\vec{B}\cdot d\vec{a} = \int_S \mu_0 \vec{j}\cdot d\vec{a}$$

$= \mu_0 I$ ↳ total current through S.

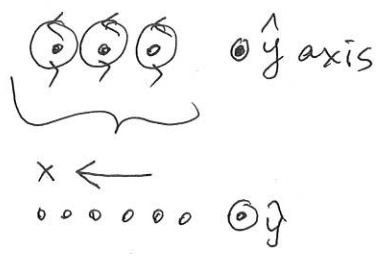
$\vec{B} = B_\theta\, \hat{\theta}$

$$\oint \vec{B}\cdot d\vec{l} = B_\theta \cdot 2\pi R = \mu_0 I$$

$$B_\theta = \frac{\mu_0 I}{2\pi R}, \quad B_r = B_z = 0$$

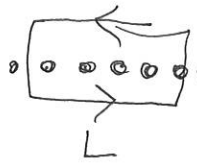


Current sheet in $z=0$ plane



Surface current density $K = \sigma v$

surface charge density



$\vec{K} = K y$, $\oint \vec{B}\cdot d\vec{l} = 2BL = \mu_0 K$

$$\vec{B}_x(z) = \tfrac{1}{2}\mu_0 K\, \text{sign}(z)\, \hat{x}.$$

Alternative representation of $\vec{B}$: magnetic potential ψ

<u>Outside the</u> region where $\vec{j} \neq 0$, $\vec{\nabla} \times \vec{B} = 0$

$\Rightarrow$ let $\vec{B} = -\vec{\nabla}\psi$

Then $\vec{\nabla} \cdot \vec{B} = 0$ gives $\nabla^2 \psi = 0$ (Laplace eqn)

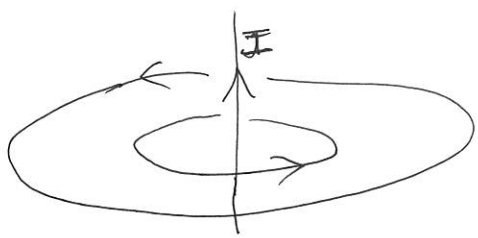
This is convenient sometimes

For a wire $\psi = -\dfrac{\mu_0 I \theta}{2\pi}$ so that $\vec{B} = -\vec{\nabla}\psi = \dfrac{\mu_0 I}{2\pi R} = B_\theta \hat{\theta}$

[in cyll. coordinates $\vec{\nabla}\psi = \dfrac{\partial \psi}{\partial \rho}\hat{\rho} + \dfrac{1}{\rho}\dfrac{\partial \psi}{\partial \theta}\hat{\theta} + \dfrac{\partial \psi}{\partial z}\hat{z}$]

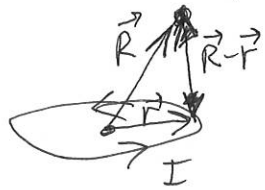
Interesting feature $\psi(\theta) = -\dfrac{\mu_0 I}{2\pi}\theta$ is multivalued:

upon $\theta \to \theta + 2\pi$ change it changes by $(-\mu_0 I)$.



That does not depend on how far from I we circle the current. Current represents line "defect" which affects potential $\to$ "topological" feature.

Consider loop current I now



$$\vec{B}_i = \left(\vec{\nabla} \times \vec{A}\right)_i = \frac{\mu_0}{4\pi}\left(\vec{\nabla}_R \times \int d^3r \, \frac{\vec{j}(\vec{r})}{|\vec{R}-\vec{r}|}\right)_i$$

$$= \frac{\mu_0 I}{4\pi} \varepsilon_{ijk} \frac{\partial}{\partial X_j} \oint \frac{dl_k}{|\vec{R}-\vec{r}|}$$

$\vec{R} = (X, Y, Z)$ used $dV \vec{j} = I \, d\vec{l}$ here

$\vec{r} = (x, y, z)$

Stokes th: $\oint_C \vec{f}\cdot d\vec{l} = \int_\Sigma \vec{\nabla}\times\vec{f}\cdot d\vec{a}$

$$\int \frac{dl_k}{|\vec{R}-\vec{r}|} = \int da_\ell \; \varepsilon_{k\ell m} \frac{\partial}{\partial x_m} \frac{1}{|\vec{R}-\vec{r}|} \quad \text{(proof below)}$$

few pages

$$B_i = \frac{\mu_0 I}{4\pi} \underbrace{\varepsilon_{ijk}\,\varepsilon_{k\ell m}}_{\delta_{i\ell}\delta_{jm}-\delta_{im}\delta_{j\ell}} \frac{\partial}{\partial X_j} \int da_\ell \frac{\partial}{\partial x_m} \frac{1}{|\vec{R}-\vec{r}|} =$$

$$= \frac{\mu_0 I}{4\pi} \left(\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}\right)\left(-\frac{\partial^2}{\partial X_j X_m}\right)\int da_\ell \frac{1}{|\vec{R}-\vec{r}|}$$

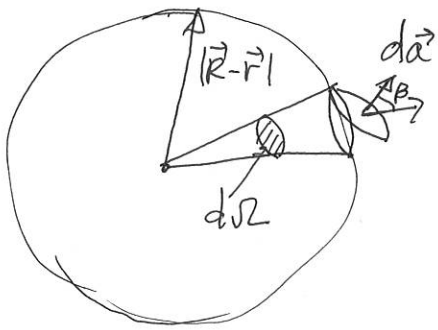
$$\delta_{jm} \frac{\partial^2}{\partial X_j^2} \frac{1}{|\vec{R}-\vec{r}|} = \nabla^2 \frac{1}{|\vec{R}-\vec{r}|} = 0 \text{ since } \vec{R} \neq \vec{r}:$$

$$\nabla^2 \frac{1}{|\vec{R}-\vec{r}|} = -4\pi\,\delta(\vec{R}-\vec{r})$$

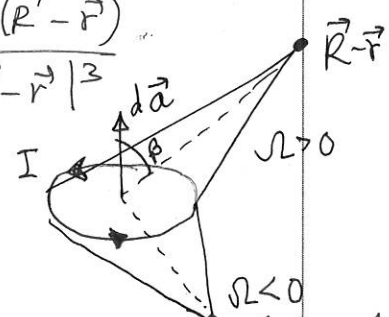
$$B_i = \frac{\mu_0 I}{4\pi} \frac{\partial^2}{\partial X_j \partial X_i} \int da_j \frac{1}{|\vec{R}-\vec{r}|} = \frac{\mu_0 I}{4\pi}\left(-\frac{\partial}{\partial X_i}\right)\int da_j \frac{(X_j - x_j)}{|\vec{R}-\vec{r}|^3}$$

$$\frac{\partial}{\partial X_j} \frac{1}{|\vec{R}-\vec{r}|} = -\frac{X_j - x_j}{|\vec{R}-\vec{r}|^3}$$

$$\Rightarrow \vec{B} = -\vec{\nabla}\psi, \quad \psi(\vec{R}) = \frac{\mu_0 I}{4\pi}\int \frac{d\vec{a}\cdot(\vec{R}-\vec{r})}{|\vec{R}-\vec{r}|^3}$$



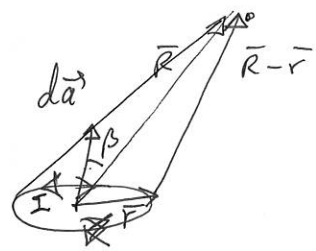
$$\hat{n} = \frac{\vec{R}-\vec{r}}{|\vec{R}-\vec{r}|}$$



$d\vec{a}\cdot\hat{n} = da\cos\beta$ area of element $d\vec{a}$ seen from point $\vec{R}-\vec{r}$.

$$\Rightarrow \quad \frac{d\vec{a}\cdot\hat{n}}{|\vec{R}-\vec{r}|^2} = d\Omega$$ solid angle that the area element $d\vec{a}$ subtends at the point $\vec{R}$

$$\Rightarrow \psi(\vec{R}) = \frac{\mu_0 I}{4\pi}\int d\Omega = \frac{\mu_0 I}{4\pi}\,\Omega(\vec{R})$$ (quite a topological expression!)



$d\Omega > 0$ when $\beta < \pi/2$

$d\Omega < 0$ $\quad \frac{\pi}{2} < \beta < \pi$

Normal $d\vec{a}$ direction of the loop with current is determined by the <u>right-hand-rule</u>.

For a small current loop of area a, $\psi(\vec{R}) = \frac{\mu_0 I}{4\pi}\frac{a\cos\theta}{R^2}$ (in spherical coordinates)

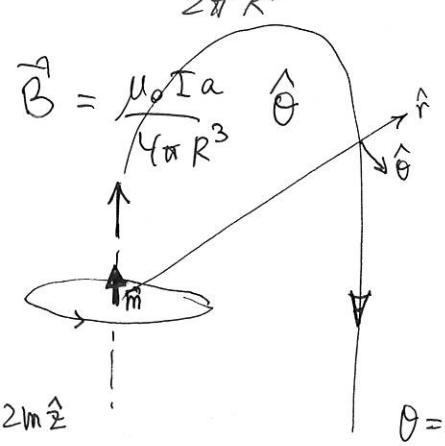
$$\vec{B}(\vec{R}) = -\vec{\nabla}_R\,\psi = -\left(\frac{\partial\psi}{\partial R}\hat{r} + \frac{1}{R}\frac{\partial\psi}{\partial\theta}\hat{\theta} + \frac{1}{R\sin\theta}\frac{\partial\psi}{\partial\varphi}\hat{\varphi}\right)$$

$$= \frac{\mu_0 I a}{4\pi}\left(\frac{2\cos\theta}{R^3}\hat{r} + \frac{\sin\theta}{R^3}\hat{\theta}\right)$$ just magnetic field of a dipole with magnetic moment $\underline{\vec{\mu} = I\vec{a}}$

For example, along the $\hat{z}$-axis $\theta = 0$

$$\vec{B} = \frac{\mu_0 I a}{2\pi R^3}\hat{z} = \frac{\mu_0 \vec{m}}{2\pi R^3}, \quad \vec{m} = I\cdot\text{Area}\,\hat{z}.$$

for $\theta = \pi/2$: $\vec{B} = \frac{\mu_0 I a}{4\pi R^3}\hat{\theta}$



Check:

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{3(\vec{m}\cdot\hat{n})\hat{n} - \vec{m}}{R^3}$$

$\theta = 0$: $\vec{m}\cdot\hat{n} = m \Rightarrow 3m\hat{z} - m\hat{z} = 2m\hat{z}$

$\theta = \frac{\pi}{2}$: $\vec{m}\cdot\hat{n} = 0$

$$\varepsilon_{ijk} \frac{\partial}{\partial X_j} \oint dl_k \frac{1}{|\vec{R}-\vec{r}|} = \varepsilon_{ijk}\varepsilon_{klm}\left(-\frac{\partial^2}{\partial X_j \partial X_m}\right) \cdot \int da_l \frac{1}{|\vec{R}-\vec{r}|}$$

Proof:

Stokes

$$\oint_C d\vec{l}\cdot\vec{A} = \int_S d\vec{a}\cdot\vec{\nabla}\times\vec{A}$$

$$\oint dl_i A_i = \int da_i\, \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j} = \int \varepsilon_{ijk}\, da_j \frac{\partial}{\partial x_k}(A_i)$$

$$= \int da_i\, \varepsilon_{ikj} \frac{\partial A_j}{\partial x_k}$$

$$= \int da_j\, \underbrace{\varepsilon_{jki}}_{\varepsilon_{ijk}} \frac{\partial A_i}{\partial x_k}$$

Thus let $A_i = \frac{1}{|\vec{R}-\vec{r}|}$

$$\oint_C dl_k \frac{1}{|\vec{R}-\vec{r}|} = \int_S \varepsilon_{klm}\, da_l \frac{\partial}{\partial x_m} \frac{1}{|\vec{R}-\vec{r}|}$$

$$\frac{\partial}{\partial x_m} = -\frac{\partial}{\partial X_m}$$

Current loop

for $z=0+$: $\Omega = 2\pi$

$$\int d\Omega = \int_0^{2\pi} d\varphi \int_0^{\theta_0} d\theta \sin\theta = 2\pi\left(-\cos\theta\right)\Big|_0^{\theta_0} = 2\pi\left(1-\cos\theta_0\right)$$

$$\sqrt{z^2+a^2}\cos\theta_0 = z$$

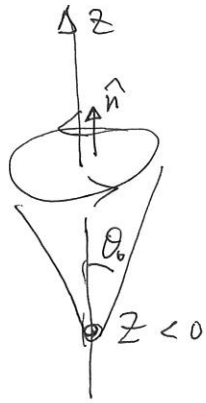
$$\psi(z>0) = \frac{1}{2}\mu_0 I\left(1-\frac{z}{\sqrt{z^2+a^2}}\right) \quad \text{for } z>0$$

For $z<0$: $\psi(z<0) = -\frac{1}{2}\mu_0 I\left(1-\frac{|z|}{\sqrt{z^2+a^2}}\right)$

since $d\Omega$ changes sign

$$= -\frac{1}{2}\mu_0 I\left(1+\frac{z}{\sqrt{z^2+a^2}}\right)$$

for $z=0-$: $\Omega = -2\pi$



below:

$$\cos\theta_0 = \frac{|z|}{\sqrt{z^2+a^2}}$$

Thus $\psi(z\to 0^+) = \frac{1}{2}\mu_0 I$

$\psi(z\to 0^-) = -\frac{1}{2}\mu_0 I$

$\Big\}\ \Delta\psi = \mu_0 I$

the potential ψ (and Ω too) is discontinuous at $z=0$.

But the field

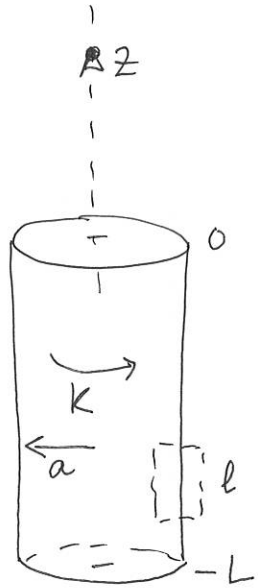
$$\vec{B} = -\frac{\partial}{\partial z}\psi\,\hat{z} = \hat{z}\,\frac{\mu_0 I a^2}{2\left(z^2+a^2\right)^{3/2}}$$

is continuous.

$$\vec{B} \to \frac{1}{2}\frac{\mu_0 I a^2}{z^3}\hat{z} = \frac{1}{2\pi}\frac{\mu_0 \vec{m}}{z^3}; \quad \underline{\vec{m} = \pi a^2 I \hat{z}}\,!$$

Solenoid

$$\psi(\vec{R}) = \frac{\mu_0 I}{4\pi} \int d\vec{a} \cdot \frac{\vec{R} - \vec{r}}{|\vec{R} - \vec{r}|^3}$$



$K = nI$ (n turns per unit length)

Field inside $B_0 \cancel{l} = \mu_0 n I \cancel{l}$

$B_0 = \mu_0 n I$

Use superposition: $\psi = \sum_{loop} \psi_{loop}$

$\vec{R} = (\rho, 0, z)$

$\vec{r} = (x, y, z)$

$d\vec{a} = (0, 0, dx\,dy)$

$$\psi(\rho, z) = \int_{-L}^{0} dz \, \frac{\mu_0 n I}{4\pi} \cdot$$

$$\cdot \int_{x^2+y^2 \le a^2} dx\,dy \, \frac{Z - z}{\left\{(\rho - x)^2 + y^2 + (Z - z)^2\right\}^{3/2}}$$

$$= \frac{B_0}{4\pi} \int dx\,dy \int_{-L}^{0} dz \, \frac{\partial}{\partial z} \frac{1}{\left\{(\rho - x)^2 + y^2 + (Z - z)^2\right\}^{1/2}}$$

$$= \frac{B_0}{4\pi} \int dx\,dy \left\{ \frac{1}{\sqrt{(\rho - x)^2 + y^2 + Z^2}} - \frac{1}{\sqrt{(\rho - x)^2 + y^2 + (Z + L)^2}} \right\}$$

At large distances $Z \gg a$, and taking $L \to \infty$

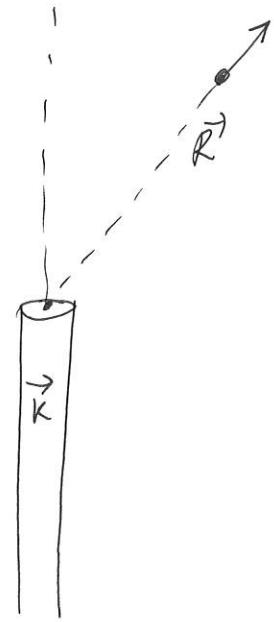
$\int dx\,dy = \pi a^2$ (area)

$$\psi(\rho, z) \approx \frac{B_0}{4\pi} \frac{1}{\sqrt{\rho^2+z^2}} \underbrace{\int dx\,dy}_{\pi a^2} = \frac{B_0 a^2}{4R}$$

Valid when $a < R = \sqrt{\rho^2+z^2} < L$

(for $R \gg L$ the solenoid turns into current loop)

$$\vec{B}(\vec{R}) = -\vec{\nabla}_R\, \psi(\vec{R}) = \frac{B_0 a^2}{4R^3}\, \vec{R}$$



the field is <u>radial.</u>

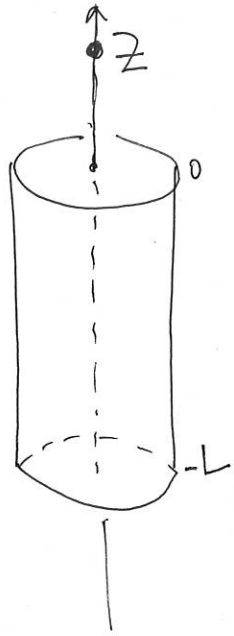
Same form as $\vec{E}$ field of electric charge $\vec{E} = \frac{q\vec{R}}{4\pi\varepsilon_0 R^3}$

$\Rightarrow$ magnetic monopole of "charge" $(B_0 \pi a^2)$

(Dirac)

Magnetic potential of the solenoid.

$$\psi(\rho,z) = \frac{B_0}{4\pi}\int dx\,dy \int_{-L}^{0} dz \frac{z-z}{\left[(\rho-x)^2+y^2+(z-z)^2\right]^{3/2}}$$



$$\int_{-L}^{0} dz \frac{\partial}{\partial z} \frac{1}{\left[(\rho-x)^2+y^2+(z-z)^2\right]^{1/2}} =$$

$$= \frac{1}{\sqrt{(\rho-x)^2+y^2+z^2}} - \frac{1}{\sqrt{(\rho-x)^2+y^2+(z+L)^2}}$$

$$B_0 = \mu_0 K = \mu_0 n I$$

$$\psi(\rho,z) = \frac{B_0}{4\pi}\int_{r^2<a^2} dx\,dy \left(\frac{1}{\sqrt{(\rho-x)^2+y^2+z^2}} - \frac{1}{\sqrt{(\rho-x)^2+y^2+(z+L)^2}}\right)$$

For point on the axis, $\rho=0$, and for $z \gg a$:

$$\psi(0,z) \cong \frac{B_0}{4\pi}\underbrace{\int dx\,dy}_{\pi a^2}\left(\frac{1}{\sqrt{z^2}} - \frac{1}{\sqrt{(z+L)^2}}\right)$$

$$= \frac{B_0 a^2}{4}\left(\frac{1}{z} - \frac{1}{z+L}\right)$$

$$\vec{B} = -\vec{\nabla}\psi = -\frac{\partial}{\partial z}\psi\,\hat{z} = \frac{B_0 a^2}{4}\left(\frac{1}{z^2} - \frac{1}{(z+L)^2}\right)\hat{z}$$

is the field of "magnetic" dipole:

$$\vec{B} \simeq \frac{B_0 a^2}{4}\frac{2L}{z^3}\,\hat{z} \quad (\text{for } z \gg L) \sim \frac{1}{z^3}$$

far away from the solenoid.